

9. [Maximum mark: 8]

Consider the homogeneous differential equation  $\frac{dy}{dx} = \frac{x-y}{x+y}$ , where  $x > 0$  and  $y \neq -x$ .

It is given that  $y = 0$  when  $x = 2$ .

By using the substitution  $y = vx$ , show that the solution of the differential equation is  $x^2 - 2xy - y^2 = 4$ .

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